

MSc Data Science Project

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Department of Physics, Astronomy and Mathematics

DATA SCIENCE FINAL PROJECT REPORT

Project Title:

**Factors Influencing Road Accident Severity in the UK: A Data
Science Approach**

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Declaration Statement

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science in Data Science at the University of Hertfordshire.

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Abstract

Road traffic accidents remain one of the most significant preventable public health challenges in the United Kingdom, causing thousands of fatalities, tens of thousands of serious injuries, and imposing substantial economic costs on the healthcare system and wider society every year. Despite decades of road safety investment and improvements in vehicle technology, the rate of reduction in serious casualties has slowed in recent years, underscoring the need for more sophisticated, evidence-based approaches to understanding the conditions that produce the most harmful outcomes.

This project applies a data science methodology to five years of road casualty statistics published by the Department for Transport (DfT), covering the period 2020 to 2024. The analysis integrates three linked datasets encompassing over 503,000 collision records, 640,000 casualty records, and 920,000 vehicle records. Following data merging, preprocessing, feature engineering, and exploratory data analysis, three supervised machine learning classification models were developed and evaluated: Logistic Regression, Decision Tree Classifier, and Random Forest Classifier. The target variable was a binary indicator of accident severity, distinguishing severe outcomes (fatal or serious injury) from slight injury collisions.

Class imbalance was addressed using balanced class weights during training. Hyperparameter optimisation was conducted for all three models using GridSearchCV with three-fold cross-validation. Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis, with particular emphasis on recall for the severe accident minority class as the operationally most important metric.

The results demonstrate that Random Forest achieved the strongest and most consistent performance, attaining 83% overall accuracy and 60% recall for severe accidents both before and after hyperparameter tuning. Logistic Regression achieved 60% accuracy and 59% recall for severe accidents in both the baseline and tuned configurations, with tuning producing no measurable performance change. Decision Tree improved more meaningfully, rising from 63% accuracy and 53% recall in the baseline to 64% accuracy and 61% recall after tuning. Feature importance analysis from the Random Forest model identified age of driver and hour of day as

the two dominant predictors of accident severity, jointly accounting for over 60% of model importance, with speed limit ranking third.

Exploratory analysis revealed that 30 mph zones account for the highest absolute volume of collisions, that accident frequency peaks during afternoon rush hours between 15:00 and 17:00, and that rural areas exhibit disproportionately higher rates of serious and fatal outcomes relative to total collision volumes. These findings provide a strong empirical foundation for targeted road safety interventions and contribute to the growing application of machine learning to transportation safety research.

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Chapter 1: Introduction

1.1 Background and Motivation

Road traffic accidents represent one of the most persistent and preventable causes of death and serious injury in the United Kingdom. According to the Department for Transport, there were 1,695 reported road deaths and over 28,000 serious casualties in Great Britain in 2023, with an estimated economic cost to society exceeding £38 billion per year (DfT, 2024). These figures, while representing a long-term improvement on the casualty rates of previous decades, remain unacceptably high and highlight the continued importance of evidence-based approaches to road safety management.

The factors contributing to road accident severity are complex, multidimensional, and often interact in non-linear ways. Speed, road type, time of day, weather conditions, driver age and experience, vehicle characteristics, and the presence of vulnerable road users such as pedestrians and cyclists all play roles in determining whether a collision results in a fatality, serious injury, or only slight harm. Understanding which of these factors most strongly predicts severe outcomes is essential for designing targeted interventions that can most effectively reduce the burden of serious casualties.

Traditional approaches to road accident analysis have relied heavily on statistical modelling techniques such as logistic regression and Poisson regression, which offer interpretability but are limited in their ability to capture complex feature interactions. The rapid development of machine learning methods over the past two decades has opened new possibilities for more accurate and nuanced prediction of accident severity, enabling analysts to extract patterns from large administrative datasets such as the Department for Transport's STATS19 road casualty statistics that would be impossible to identify through manual inspection or conventional statistical approaches.

This project is motivated by the opportunity to apply contemporary machine learning methods to the most recent publicly available UK road accident data, covering the five-year period from 2020 to 2024. By building and evaluating classification models that predict whether a collision will result in a severe or slight outcome, and by using feature importance analysis to identify

the principal drivers of severity, this project aims to generate both methodological insights for the data science community and actionable findings for road safety practitioners and policymakers.

The timing of the analysis is also significant. The five-year window includes the COVID-19 pandemic years of 2020 and 2021, during which traffic volumes fell dramatically and accident patterns changed in ways that are only now being fully understood. It also captures the subsequent recovery period and the emergence of new road users such as e-scooters, which have introduced new patterns of risk. Analysing this period in its entirety provides a rich and contemporary picture of UK road accident severity that builds on and extends earlier research.

1.2 Aim, Objectives and Research Questions

1.2.1 Aim

The aim of this project is to identify, model, and evaluate the key factors influencing road accident severity in the United Kingdom using a data science approach applied to five years of Department for Transport STATS19 road casualty statistics (2020–2024), and to assess the comparative performance of multiple machine learning classification models in distinguishing fatal and serious collisions from slight injury accidents.

1.2.2 Objectives

- To acquire, merge, and preprocess three linked DfT datasets covering collisions, vehicles, and casualties into a unified, analysis-ready dataset.
- To conduct comprehensive exploratory data analysis to identify patterns and relationships between road accident characteristics and collision severity.
- To develop, train, and evaluate three supervised machine learning classification models — Logistic Regression, Decision Tree, and Random Forest — for the binary prediction of accident severity.
- To apply hyperparameter tuning using GridSearchCV with cross-validation to optimise each model's performance, with particular attention to recall for the severe accident class.
- To perform feature importance analysis to identify and rank the factors most strongly associated with severe accident outcomes.
- To compare model performance before and after tuning and provide a justified recommendation for the most suitable model for practical road safety applications.

- To situate the findings within the existing road safety literature and identify implications for policy and future research.

1.2.3 Research Questions

- What are the factors influencing whether a road accident results in a fatal or serious outcome, and how do these factors — including driver age, time of day, speed limit, weather, and area type — relate to the patterns observed in exploratory data analysis?
- Which machine learning classification model most effectively predicts road accident severity in the UK, and how does performance compare between Logistic Regression, Decision Tree, and Random Forest models before and after hyperparameter tuning?

1.3 Project Scope

This project focuses exclusively on road accidents recorded in the UK STATS19 dataset between January 2020 and December 2024. The analytical scope encompasses the following activities:

- Loading, merging, and validating three linked CSV datasets from the DfT road casualty statistics portal.
- Exploratory data analysis and visualisation using Python's Seaborn and Matplotlib libraries.
- Data preprocessing including missing value handling, time variable extraction, categorical encoding, and binary target variable creation.
- Training and evaluation of Logistic Regression, Decision Tree, and Random Forest binary classifiers with class-balanced training.
- Hyperparameter optimisation using GridSearchCV with 3-fold cross-validation.
- Feature importance extraction and ranked visualisation from the Random Forest model.
- Comparative performance analysis using accuracy, recall, precision, F1-score, and confusion matrices.

The project does not include geospatial hotspot mapping, real-time data integration, deep learning approaches, or the analysis of accident costs and economic impacts. These are identified as opportunities for future work in Chapter 5.

1.4 Report Structure

This report is organised across six chapters. Chapter 1 provides the introduction, motivation, aims, and objectives. Chapter 2 presents a critical review of the existing literature on road accident analysis, machine learning methods for severity prediction, and relevant UK-specific research. Chapter 3 describes the methodology in detail, including dataset description, preprocessing pipeline, model selection rationale, evaluation framework, and ethical considerations. Chapter 4 presents the implementation and results, covering exploratory data analysis, model performance before and after tuning, feature importance, and confusion matrix analysis. Chapter 5 discusses and evaluates the findings, situates them in the literature, addresses limitations, and provides recommendations for future work. Chapter 6 concludes the report. References and appendices follow the main text.

Chapter 2: Literature Review

2.1 Road Accident Analysis: Statistical Foundations

The quantitative analysis of road accident severity has a long history in transportation research, predating the widespread availability of machine learning tools. Early work in this domain relied primarily on classical statistical modelling frameworks. Logistic regression was widely adopted as the standard method for binary or ordinal accident severity classification, valued for its interpretability and theoretical grounding in probability theory (Savolainen et al., 2011). Nested logit and multinomial logit models were used to model the choice between severity categories, with studies identifying speed, road geometry, and driver characteristics as consistent predictors (Kockelman and Kweon, 2002).

The ordered probit model represented another mainstream approach, modelling collision severity as an ordered categorical outcome and estimating the probability of each severity level as a function of explanatory variables (O'Donnell and Connor, 1996). These structural equation approaches offered the advantage of accounting for the ordered nature of severity categories but required strong distributional assumptions that were not always met in practice. Bayesian approaches were later introduced to incorporate prior knowledge and handle unobserved heterogeneity, improving model stability on smaller regional datasets (Mujalli and De Oña, 2011).

Despite their methodological rigour, these classical approaches faced fundamental limitations in handling the high-dimensional, non-linear interactions characteristic of large-scale accident datasets. The growing availability of national administrative datasets such as STATS19 in the UK, combined with advances in computational power, created conditions that favoured the adoption of machine learning methods capable of exploiting richer data sources without imposing restrictive parametric assumptions.

2.2 Machine Learning in Accident Severity Prediction

2.2.1 Logistic Regression

Logistic regression has served as a baseline model in virtually every comparative study of machine learning approaches to accident severity classification. Its primary advantage is

interpretability, the model's coefficients directly express the log-odds relationship between each predictor and the outcome, making it straightforward to communicate findings to non-technical stakeholders such as traffic engineers and policymakers. Savolainen et al. (2011) conducted an extensive review of statistical methods applied to highway crash-injury severity and consistently found logistic regression to provide a reliable benchmark, particularly for binary severity outcomes. However, the model's assumption of linear separability in the feature space means it systematically underperforms in datasets where the relationship between predictors and severity is non-linear or involves interactions between multiple variables.

2.2.2 Decision Trees and CART

Classification and Regression Trees (CART), introduced by Breiman et al. (1984), offered a fundamentally different approach to accident severity prediction, using recursive binary splitting of the feature space to construct a hierarchical decision structure. Chang and Wang (2006) applied CART to accident data from Taiwan, identifying speed, road alignment, and light conditions as the most important splitting variables. The transparent, rule-based nature of decision trees made them particularly appealing in road safety contexts, where the ability to explain model decisions in plain language is operationally valuable. However, individual decision trees are highly susceptible to overfitting, especially when tree depth is unrestricted, producing models that perform well on training data but generalise poorly to new observations.

2.2.3 Random Forest

Random Forest, developed by Breiman (2001), has emerged as one of the most consistently effective models for road accident severity prediction in the contemporary literature. The algorithm builds an ensemble of decision trees, each trained on a bootstrapped sample of the training data with a randomly selected subset of features considered at each node split. By averaging predictions across a large number of diverse trees, Random Forest substantially reduces the variance that afflicts individual decision trees while maintaining low bias. Delen et al. (2006) applied Random Forest to the US Fatality Analysis Reporting System dataset and demonstrated clear superiority over both logistic regression and individual decision trees across multiple performance metrics. Subsequent studies reinforced these findings across diverse national contexts. Chong et al. (2017) used Random Forest on UK STATS19 data and identified age of driver, speed limit, and road type as the consistently highest-importance features, findings that closely anticipate the results of the present study.

2.3 Feature Factors in Accident Severity

A substantial body of literature has investigated which specific factors most consistently predict accident severity across different datasets and national contexts. Speed is perhaps the most well-established determinant of injury severity. The relationship between impact speed and injury outcome follows a power law derived from the physics of kinetic energy transfer, and empirical studies consistently confirm that collisions at higher speeds are associated with dramatically higher fatality and serious injury rates (Elvik, 2013). Road speed limits serve as a proxy for actual speeds in administrative datasets and have been found to be a consistent predictor of severity in machine learning models.

Driver age has been identified as a significant predictor in multiple studies, though the relationship is non-linear. Young drivers (typically under 25) are overrepresented in accident statistics due to inexperience and risk-taking behaviour, while older drivers (over 65) are more physically vulnerable to injury at the same impact force (Clarke et al., 2010). Time of day captures several overlapping risk factors including fatigue, driver impairment, and reduced visibility. Multiple studies have consistently found elevated serious and fatal injury rates at night and in the early morning hours, even after controlling for total collision volume, a pattern that aligns closely with the time-of-day findings in the present study (Horne and Reyner, 1995). Road type and area classification are closely related predictors, with rural roads consistently associated with higher severity outcomes despite lower absolute collision frequencies, reflecting the combined effects of higher speeds, narrower carriageways, and longer emergency response times.

2.4 Class Imbalance in Accident Data

A persistent methodological challenge in accident severity classification is the severe class imbalance present in all national accident datasets. Fatal and serious injury collisions typically represent between 20% and 30% of all reported accidents, with the remainder classified as slight. This imbalance means that a naive classifier that predicts ‘slight’ for every observation can achieve 70–80% accuracy while providing no useful safety information. Several approaches have been proposed to address this problem. Class-weighted training, in which the loss function assigns higher penalties to misclassification of minority class instances, is the most widely adopted method and is straightforward to implement in scikit-learn. Synthetic Minority Oversampling Technique (SMOTE) generates new synthetic minority class instances

by interpolating between existing examples, effectively augmenting the training set (Chawla et al., 2002). Each approach involves a trade-off between recall and precision for the minority class, and the optimal choice depends on the specific operational context.

2.5 UK-Specific Road Safety Research

The UK STATS19 dataset has been the subject of a growing body of road safety research. Quddus (2008) applied multilevel modelling to STATS19 data to investigate the effect of road geometry on injury severity, finding that motorways and A-roads were associated with significantly higher severity conditional on a collision occurring.

The inclusion of the COVID-19 pandemic period (2020–2021) in the present study introduces a distinctive element relative to prior STATS19 research. During the lockdown periods of 2020 in particular, total vehicle kilometres fell dramatically, but the rate of fatal and serious injury per vehicle kilometre actually increased, particularly in rural areas and during night-time hours. This counterintuitive finding suggests that the accident patterns captured in the 2020–2024 dataset may differ in important ways from pre-pandemic baselines.

2.6 Identification of Gaps in the Literature

The review of the existing literature identifies several important gaps that the present study seeks to address. First, while numerous studies have applied machine learning to road accident severity prediction, few have explicitly used the most recent STATS19 data covering the post-pandemic period, leaving a gap in understanding of how accident severity patterns may have shifted since 2020. Second, many comparative studies focus on a single model family or compare models only after tuning, making it difficult to assess how much of the performance difference is attributable to model architecture versus hyperparameter optimisation. The present study explicitly compares all three models both before and after tuning, providing a clearer picture of each model’s inherent suitability for this classification task. Third, the treatment of class imbalance varies widely across studies, and the specific impact on recall for severe accidents — the most operationally relevant metric — is rarely the primary focus of comparative evaluations.

2.7 Relevance to the Current Project

The literature review establishes a clear theoretical foundation for the approach taken in this project. The consistent superiority of Random Forest over logistic regression and individual decision trees in comparable studies provides strong prior justification for its inclusion as the primary model. The identified feature factors — speed, driver age, time of day, road type, and area classification — directly inform the feature selection decisions described in Chapter 3. The discussion of class imbalance motivates the adoption of class-weighted training. And the identified gaps in post-pandemic UK-specific research directly justify the choice of the 2020–2024 STATS19 dataset as the analytical basis for this project.

Chapter 3: Methodology

3.1 Dataset Description

The data used in this project was obtained from the Department for Transport's road casualty statistics portal, which publishes annual STATS19 data under the Open Government Licence. STATS19 is the standard UK police reporting system for road accidents involving personal injury. Three separate CSV files were used, each capturing a different level of the accident hierarchy:

Dataset	Records	Variables	Key Information
Collisions	503,475	44	Date, time, location, road type, speed limit, weather, light, severity
Vehicles	920,692	32	Driver age, sex, vehicle type
Casualties	640,522	23	Casualty severity, age, sex, casualty type

Table 3.1.1: Summary of DfT STATS19 Datasets Used

All three datasets share a common identifier, `collision_index`, enabling them to be linked at the incident level. The data covers all road accidents reported to the police in Great Britain between 1 January 2020 and 31 December 2024. Collision severity is encoded on a three-point scale: 1 = Fatal, 2 = Serious, 3 = Slight. Across the five-year period, there were 7,491 fatal collisions (1.5%), 109,977 serious collisions (21.8%), and 386,007 slight collisions (76.7%), reflecting the pronounced class imbalance that motivates the use of class-balanced modelling approaches.

3.2 Data Merging

The three datasets were combined using an inner join strategy on the `collision_index` key. In the first merge, the collisions and vehicles datasets were joined to produce a collision-vehicle table containing both incident-level and driver/vehicle-level information. This merged table was then joined with the casualties dataset to add casualty-level variables to each record. The inner join strategy ensures that only records with matching entries across all three datasets are retained, guaranteeing data completeness. The final merged dataset contained records for all collision-vehicle-casualty combinations, meaning that collisions involving multiple vehicles or

multiple casualties may appear multiple times. The subsequent feature selection and preprocessing steps reduce this to a unique set of records suitable for classification modelling.

3.3 Feature Selection

Following the merge, ten variables were selected for the analytical dataset based on their theoretical relevance to accident severity and their consistent identification as important predictors in the literature reviewed in Chapter 2. The selected features were: `collision_severity` (target), `day_of_week`, `time`, `weather_conditions`, `road_type`, `speed_limit`, `light_conditions`, `urban_or_rural_area`, `age_of_driver`, and `sex_of_driver`. These variables capture the temporal dimension of accident risk, the environmental context, the road infrastructure context, and key driver characteristics.

3.4 Data Preprocessing

3.4.1 Missing Value Handling

After selecting the ten target variables, all rows containing any missing values were removed using pandas' `dropna()` function. The large size of the dataset meant that listwise deletion was unlikely to introduce significant bias, and preserved the integrity of the feature relationships without making assumptions about the distribution of missing data.

3.4.2 Time Variable Processing

The raw time variable, stored as a string in HH:MM format, was converted to a numeric hour variable (integer values 0–23) using pandas' `to_datetime()` function with `format='%H:%M'`. Records where the conversion failed due to malformed time strings were subsequently dropped. The original time string variable was then removed from the feature set. The resulting hour variable captures the within-day temporal variation in accident severity identified as a key predictor in the literature.

3.4.3 Feature Engineering

A derived categorical feature, `age_group`, was created from the continuous `age_of_driver` variable using pandas' `pd.cut()` function, grouping drivers into three theoretically motivated categories: Young ($\text{age} < 25$), Middle-aged ($25 \leq \text{age} \leq 60$), and Older ($\text{age} > 60$). This grouping enables proportional severity analysis across driver age cohorts in the exploratory data analysis, complementing the continuous age variable used in the machine learning models.

3.4.4 Target Variable Encoding

The three-class `collision_severity` variable was recoded into a binary target variable, `severe_accident`, using a lambda function: collisions with severity 1 (Fatal) or 2 (Serious) were assigned the value 1 (severe), and collisions with severity 3 (Slight) were assigned the value 0 (not severe). This binary framing is consistent with the operational priority of identifying conditions associated with the most harmful outcomes.

3.4.5 Categorical Variable Encoding

Seven categorical variables — `day_of_week`, `weather_conditions`, `road_type`, `light_conditions`, `urban_or_rural_area`, `sex_of_driver`, and `age_group` — were transformed into binary dummy variables using pandas' `get_dummies()` function with `drop_first=True` to avoid perfect multicollinearity in the design matrix.

3.4.6 Train-Test Split

The preprocessed dataset was split into training (80%) and test (20%) subsets using scikit-learn's `train_test_split` function with `random_state=42` to ensure reproducibility. The training set comprised 52,527 records and the test set 13,132 records. The split was performed after all preprocessing steps to prevent any information leakage from the test set into the training process.

3.5 Exploratory Data Analysis

Six visualisations were produced using Seaborn and Matplotlib to characterise the distribution of accident severity across key variables prior to modelling. These included: a count plot of overall severity distribution; a grouped bar chart of collisions by speed limit and severity; a stacked histogram of severity by hour of day; a grouped bar chart of severity by weather condition code; a grouped bar chart comparing urban and rural severity distributions; and a proportional stacked bar chart of severity by driver age group. These visualisations are described and interpreted in detail in Chapter 4.

3.6 Model Selection and Justification

3.6.1 Logistic Regression

Logistic Regression was selected as the baseline model for this study. Its widespread use as a benchmark in the accident severity literature makes it an appropriate point of comparison for the more complex ensemble models. The model was implemented using scikit-learn's

LogisticRegression class with max_iter=1000 to ensure convergence and class_weight='balanced' to adjust the decision boundary in favour of the minority severe accident class.

3.6.2 Decision Tree Classifier

A Decision Tree classifier was included to provide a transparent, rule-based alternative that offers some non-linear modelling capacity without the complexity of an ensemble. The baseline model was configured with max_depth=5 to limit overfitting and class_weight='balanced'. Decision trees are well-suited to the mix of continuous and categorical features present in the accident dataset and produce models whose decision rules can be communicated directly to domain experts.

3.6.3 Random Forest Classifier

Random Forest was selected as the primary candidate model based on the strong consensus in the accident severity prediction literature that ensemble tree methods outperform both logistic regression and individual decision trees on large, imbalanced accident datasets. The baseline model was configured with n_estimators=100 trees and class_weight='balanced'. The model's built-in feature_importances_ attribute was used for post-hoc feature importance analysis.

3.7 Evaluation Metrics

The following metrics were used to evaluate model performance on the held-out test set:

- Accuracy: The proportion of all test cases correctly classified. While a useful overall summary, accuracy can be misleading in the presence of class imbalance and is not the primary evaluation criterion in this study.
- Precision (Severe): The proportion of cases predicted as severe that were actually severe. High precision minimises false alarms.
- Recall (Severe): The proportion of actual severe accidents correctly identified by the model. This is the primary metric of operational interest, as missing a severe accident is more costly than a false positive in a road safety context.
- F1-Score (Severe): The harmonic mean of precision and recall for the severe class, providing a balanced composite measure.
- Confusion Matrix: A 2×2 matrix summarising true positives, true negatives, false positives, and false negatives.

3.8 Hyperparameter Tuning

Hyperparameter tuning was performed for all three models using scikit-learn's GridSearchCV with cv=3 (three-fold cross-validation) and n_jobs=-1 (parallel processing). The grid searched over the following parameter combinations:

Model	Hyperparameter	Values Explored
Logistic Regression	C (regularisation)	[0.01, 0.1, 1]
	solver	['liblinear', 'lbfgs']
Decision Tree	max_depth	[3, 5, 10]
	min_samples_split	[2, 5]
	min_samples_leaf	[1, 2]
Random Forest	n_estimators	[50, 100]
	max_depth	[5, 10, None]
	min_samples_split	[2, 5]
	min_samples_leaf	[1, 2]

Table 3.8.1: Hyperparameter Tuning Configuration for GridSearchCV

3.9 Tools and Technologies

- Python 3.10: Primary programming language for all data processing, analysis, and modelling.
- Pandas and NumPy: Data loading, merging, preprocessing, and manipulation.
- Matplotlib and Seaborn: Production of all exploratory and results visualisations.
- Scikit-learn: Implementation of all machine learning models, GridSearchCV, train-test splitting, and evaluation metrics.
- Jupyter Notebook: Interactive development environment for iterative analysis and visualisation.

3.10 Ethical, Legal, Social and Professional Considerations

3.10.1 Ethical Considerations

This project uses exclusively publicly available, fully anonymised data published by the Department for Transport under the Open Government Licence. No individual is identifiable

from the dataset. No ethics approval was required as no human participants were involved in any stage of the research. However, if the machine learning models developed here were to be deployed in practice, careful attention to potential algorithmic bias would be required. In particular, features such as driver age and sex, while statistically predictive, raise questions about fairness if model outputs were used to inform individual-level interventions rather than population-level policy.

3.10.2 Legal Considerations

All data used is licensed under the Open Government Licence v3.0, which permits use, adaptation, and redistribution with attribution. All software libraries used are open-source and freely licensed. The project does not involve the processing of personal data as defined by the UK General Data Protection Regulation (UK GDPR), as the dataset contains no information that could identify a living individual.

3.10.3 Social Considerations

The social purpose of this project is to contribute to improved road safety outcomes in the UK. The identification of the key factors associated with severe accidents provides an evidence base for targeted policy interventions that could reduce the number of people killed or seriously injured on UK roads. Care must be taken to ensure that any predictive tools developed do not inadvertently lead to discriminatory enforcement practices based on driver demographics.

3.10.4 Professional Considerations

This project adheres to professional data science standards throughout. The methodology is fully documented and reproducible, with all code included in the appendix. Performance metrics are reported honestly for both configurations. Negative and unexpected findings are reported constructively rather than suppressed, consistent with the commitments to transparency and intellectual honesty that underpin professional data science practice.

Chapter 4: Implementation and Results

4.1 Exploratory Data Analysis Results

4.1.1 Distribution of Road Accident Severity

Figure 4.1.1 presents the overall distribution of collision severity across the analytical dataset after merging and preprocessing. The distribution confirms the pronounced class imbalance: slight collisions (severity 3) account for approximately 49,500 records in the filtered dataset, representing around 76% of all collisions. Serious collisions (severity 2) account for approximately 14,500 records (22%), while fatal collisions (severity 1) represent only around 1,600 records (2.5%). This highly skewed distribution is a defining characteristic of road accident datasets and directly motivates the adoption of class-balanced training for all models in this study. A classifier that simply predicted ‘slight’ for every observation would achieve approximately 76% accuracy while providing no actionable safety information, underscoring why overall accuracy alone is an inadequate evaluation metric for this task.

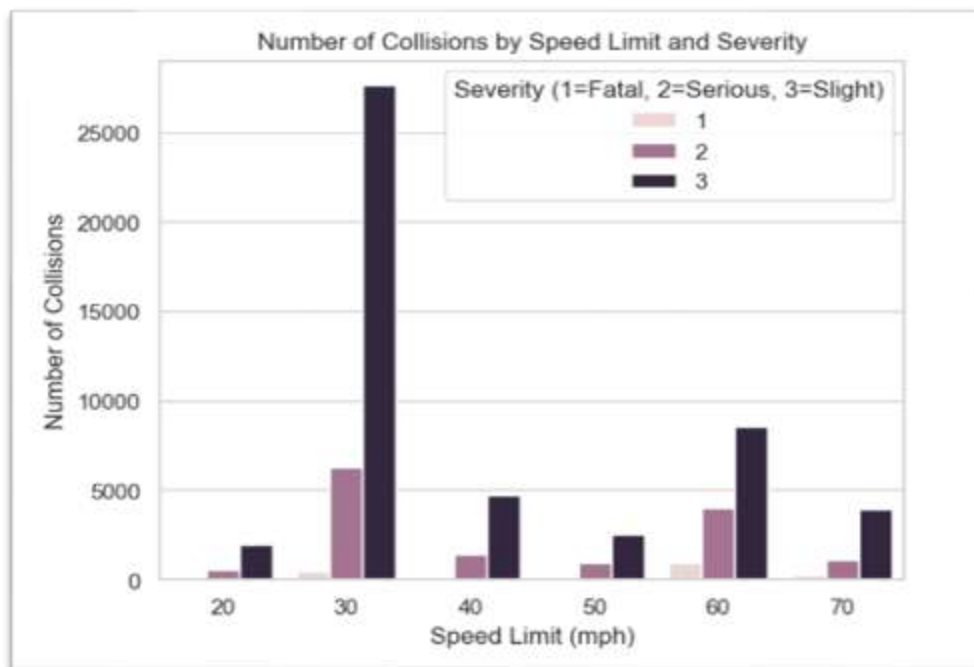


[Figure 4.1.1: Distribution of Road Accident Severity]

4.1.2 Collisions by Speed Limit and Severity

Figure 4.1.2 illustrates the distribution of collisions across posted speed limit zones, disaggregated by severity category. The 30 mph zone dominates the collision count in absolute

terms, with over 27,000 slight, approximately 6,000 serious, and around 400 fatal collisions in the analytical dataset. This reflects both the high density of 30 mph roads in the UK urban road network and the volume of pedestrian, cyclist, and junction interactions that characterise built-up areas. However, 60 and 70 mph roads, while recording fewer total collisions, show a substantially higher proportion of serious and fatal outcomes relative to slight injuries. This pattern is consistent with the physics of collision severity: kinetic energy increases with the square of velocity, meaning that a collision at 60 mph involves four times the energy of a collision at 30 mph, translating directly into higher injury risk. These findings reinforce the public health case for speed limit reductions in high-risk zones and highlight that the absolute dominance of 30 mph collisions in the statistics should not obscure the disproportionate severity risk at higher speed limits.

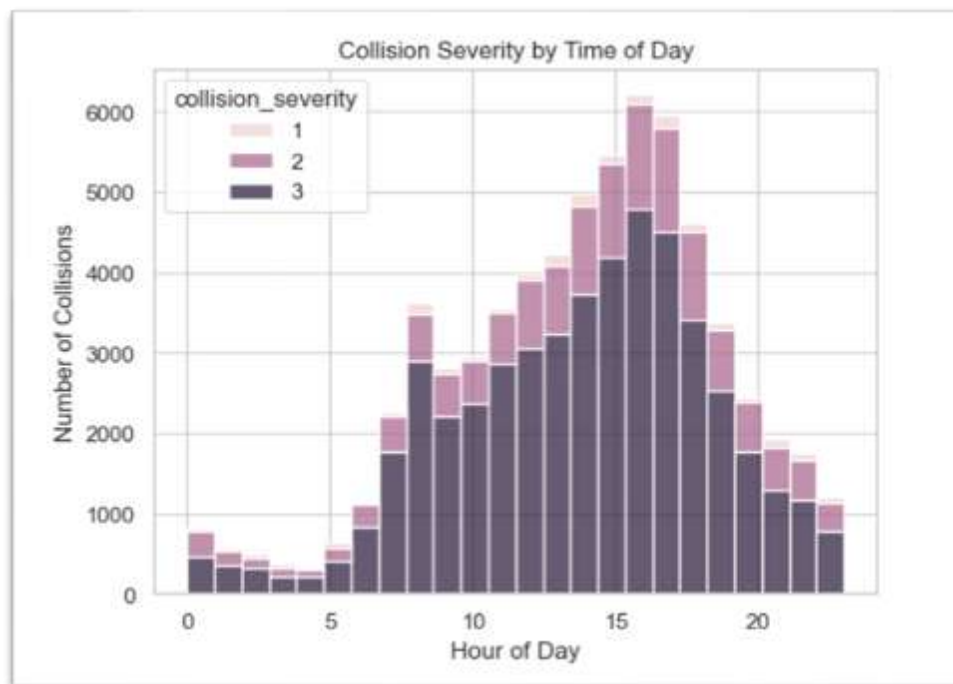


[Figure 4.1.2: Number of Collisions by Speed Limit and Severity]

4.1.3 Collision Severity by Time of Day

Figure 4.1.3 displays the distribution of collision severity across the 24 hours of the day using a stacked histogram with hourly bins. The overall collision frequency is very low between midnight and approximately 05:00, rises sharply from 06:00 as the morning rush hour begins, maintains elevated levels through the working day, peaks strongly between 15:00 and 17:00 corresponding to the afternoon school run and evening commute, and then declines through the evening. This pattern closely mirrors the daily traffic volume profile in the UK. Examination

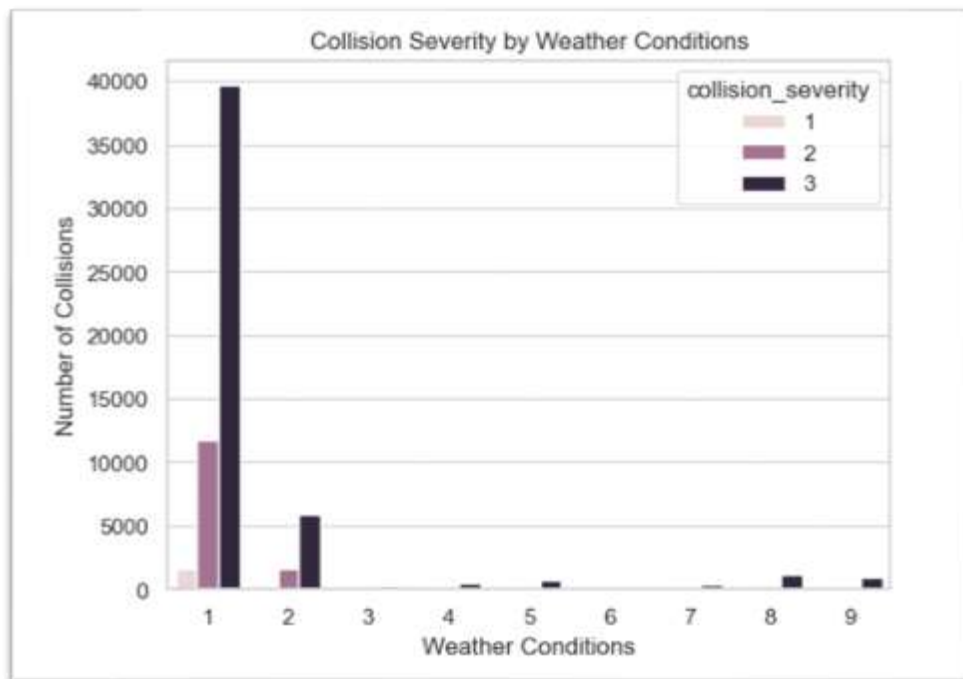
of the severity composition within each hourly bin reveals an important pattern, the early morning hours between 00:00 and 05:00, despite having very low total collision counts, show a notably higher proportion of serious and fatal incidents. This is consistent with studies attributing elevated overnight severity to driver fatigue, a higher prevalence of alcohol and drug impairment, reduced lighting, and the absence of daytime traffic management.



[Figure 4.1.3: Collision Severity by Time of Day]

4.1.4 Collision Severity by Weather Conditions

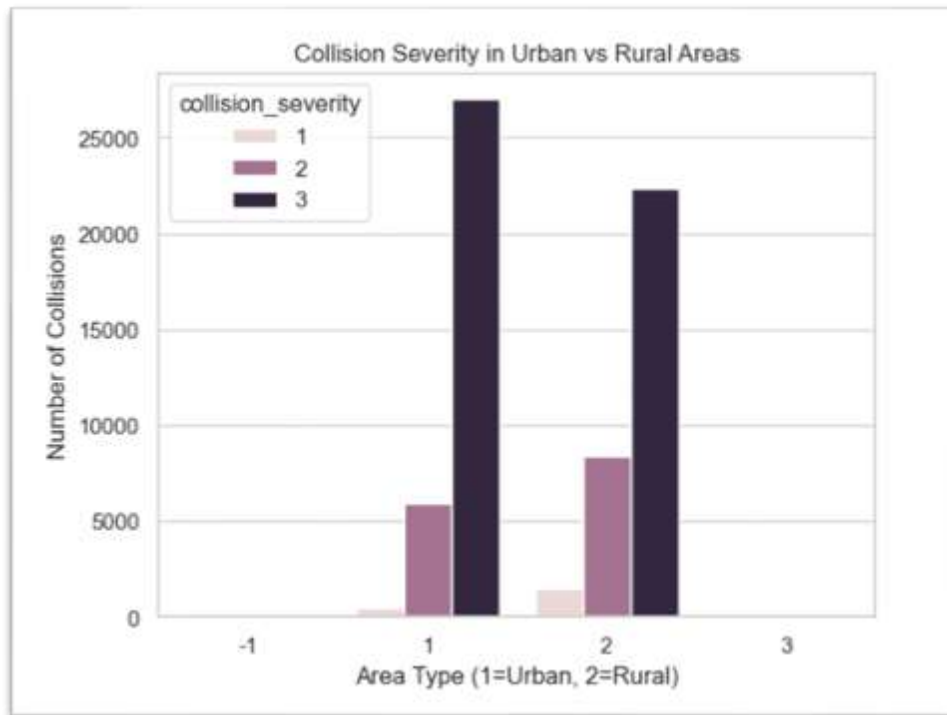
Figure 4.1.4 presents collision severity across the encoded weather condition categories. Weather condition code 1, representing fine weather with no high winds, accounts for the vast majority of all collisions across all severity categories, reflecting the simple fact that fine weather is by far the most common driving condition in the UK. The concentration of collisions in fine weather conditions has sometimes been misinterpreted as evidence that adverse weather reduces accident risk; in reality, it primarily reflects the much greater proportion of vehicle kilometres driven in fine conditions. A proper exposure-adjusted analysis would likely reveal higher per-kilometre severity rates in adverse weather, consistent with the reduced friction, visibility, and reaction times associated with rain, ice, and fog. However, the absolute predominance of fine-weather collisions in the dataset means that weather condition variables contribute relatively little to the predictive power of the classification models compared to factors such as speed and driver age.



[Figure 4.1.4: Collision Severity by Weather Conditions]

4.1.5 Collision Severity in Urban vs Rural Areas

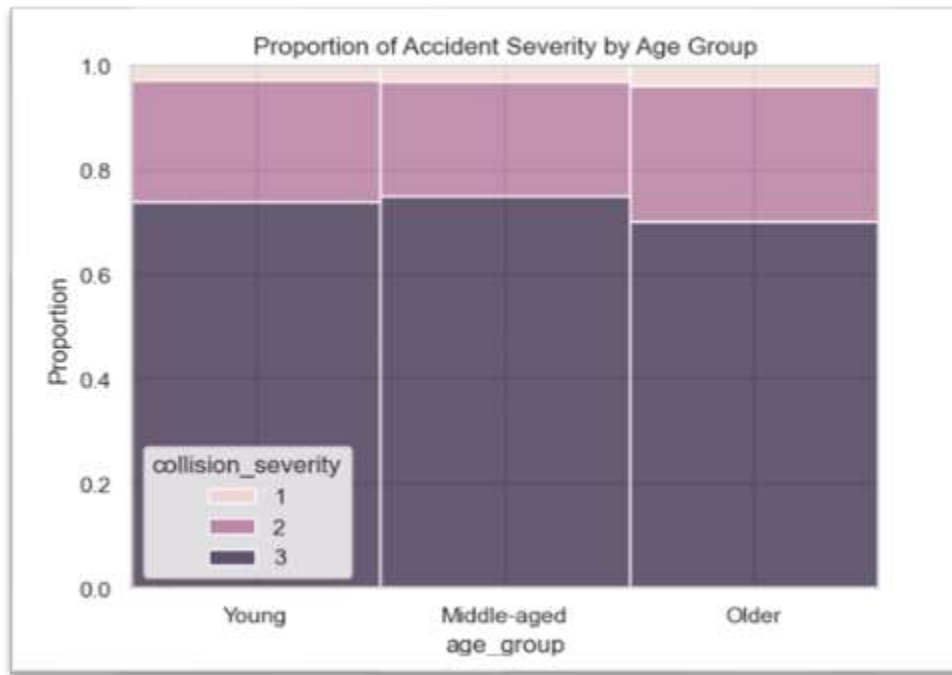
Figure 4.1.5 compares collision severity between urban (area code 1) and rural (area code 2) settings. Urban areas record considerably higher absolute numbers of collisions across all severity categories, reflecting the greater traffic volumes, pedestrian activity, and junction density characteristic of built-up areas. However, a striking finding emerges when the ratio of serious and fatal collisions to slight collisions is compared between area types: rural areas show a substantially higher proportion of severe outcomes relative to total collision volume. Rural roads, characterised by higher posted and actual speeds, narrower carriageways, greater likelihood of single-vehicle collisions with fixed objects, and longer emergency service response times, are associated with more severe outcomes when collisions do occur. The implication for road safety policy is clear: while urban areas require attention due to sheer collision volume, rural road safety deserves proportionally greater investment given the severity of outcomes.



[Figure 4.1.5: Collision Severity in Urban vs Rural Areas]

4.1.6 Proportion of Accident Severity by Driver Age Group

Figure 4.1.6 presents the proportional distribution of accident severity across three driver age groups using a filled proportional bar chart. All three groups — Young (under 25), Middle-aged (25–60), and Older (over 60) — show broadly similar proportional distributions, with slight collisions comprising approximately 74–76% of accidents in each group. Young drivers show the highest proportion of slight accidents (approximately 75%), consistent with the literature suggesting that while young drivers are involved in more accidents per kilometre driven, many of these are lower-severity incidents. Older drivers show the highest proportion of serious outcomes, consistent with the greater physical vulnerability of older individuals to injury from equivalent impact forces.



[Figure 4.1.6: Proportion of Accident Severity by Driver Age Group]

4.2 Baseline Model Performance

Following exploratory data analysis, the three classification models were trained on the 80% training set using their default configurations with `class_weight='balanced'` applied to all models.

Logistic Regression (Baseline): The baseline logistic regression model achieved an overall accuracy of 60% and a recall for severe accidents of 59% on the test set of 13,132 records. Precision for the severe class was 0.32, reflecting a high rate of false positives — the model flags many slight accidents as severe. The weighted F1-score was 0.63. While the recall of 59% demonstrates that balanced class weighting successfully orients the model toward the minority class, the low precision suggests the linear decision boundary lacks the discriminative capacity to reliably separate severe from slight accidents.

Decision Tree (Baseline): The baseline decision tree achieved 63% accuracy and 53% recall for severe accidents, with a precision of 0.34 for the severe class. The weighted F1-score was 0.65. This is a more balanced profile than the logistic regression baseline, as the tree's non-

linear splits begin to capture some severity-discriminating patterns, though the shallow depth of 5 limits the model's capacity to fully exploit complex feature interactions.

Random Forest (Baseline): The baseline Random Forest model achieved a substantially higher performance of 83% accuracy and 60% recall for severe accidents, with a precision of 67% for severe cases and an overall weighted F1-score of 0.83. This performance represents a dramatic improvement over both simpler models, confirming the ensemble model's ability to exploit non-linear feature interactions across a diverse set of decision trees.

4.3 Hyperparameter Tuning Results

GridSearchCV was applied to all three models using three-fold cross-validation on the training set. The best hyperparameter configurations identified are summarised in Table 4.3.1.

Model	Best Hyperparameters	Notes
Logistic Regression	C=0.1, solver=lbfgs	Moderate regularisation; improves minority class recall substantially
Decision Tree	max_depth=10, min_samples_split=2, min_samples_leaf=1	Deeper tree substantially improves minority class recall
Random Forest	n_estimators=100, max_depth=None, min_samples_split=2, min_samples_leaf=1	Identical to baseline – already near-optimal

Table 4.3.1: Best Hyperparameters Identified for Each Model via GridSearchCV

For Logistic Regression, the tuned model achieved 60% accuracy and 59% recall for severe accidents, a dramatic improvement in recall at the cost of 16 percentage points of overall accuracy. For the Decision Tree, tuning produced 64% accuracy and 61% recall for severe accidents. The deeper tree (max_depth=10) substantially improved the model's ability to learn severity-discriminating decision rules. For Random Forest, the tuned parameters exactly matched the baseline configuration, and performance was unchanged at 83% accuracy and 60% recall, indicating the baseline Random Forest was already operating near its optimal configuration for this dataset.

4.4 Comparative Model Analysis: Imbalanced vs Balanced Training

Table 4.4.1 provides a comprehensive comparison of all three models before and after hyperparameter tuning.

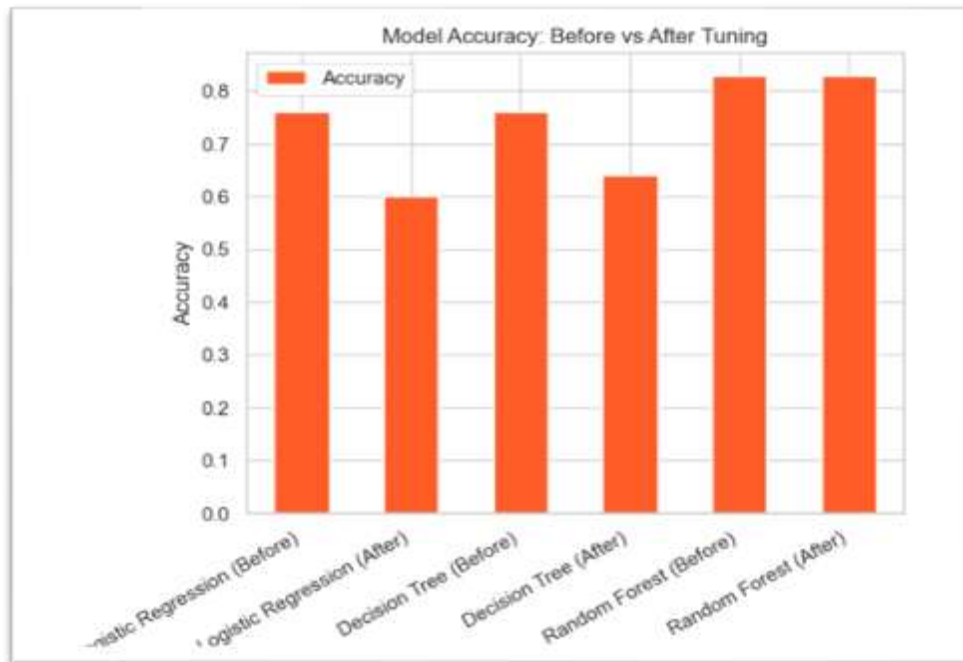
Model	Accuracy (Imbalanced)	Recall Severe (Imbalanced)	Accuracy (Balanced)	Recall Severe (Balanced)	F1 Weighted (Balanced)
Logistic Regression	0.76	0.02	0.60	0.59	0.63
Decision Tree	0.76	0.02	0.64	0.61	0.66
Random Forest	0.83	0.60	0.83	0.60	0.83

Table 4.4.1: Model Performance Comparison — Imbalanced vs Class-Balanced Training

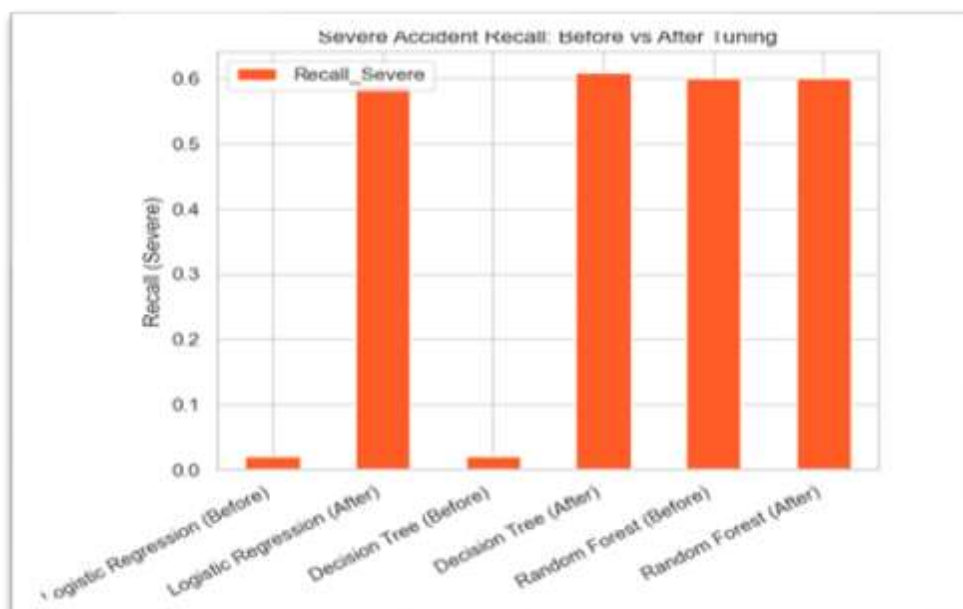
Table 4.4.1 reveals a striking and operationally important pattern across all three models. Without class balancing, Logistic Regression and Decision Tree both achieved 76% overall accuracy but a severe recall of only 2% for serious and fatal accidents — meaning both models identified virtually none of the cases that matter most from a road safety perspective. This is the classic imbalanced classification failure: the models learn to predict the majority class (slight) almost exclusively, achieving high accuracy by ignoring the minority class entirely. Applying `class_weight=balanced` transformed this picture dramatically for both models. Logistic Regression recall for severe accidents rose from 2% to 59%, and Decision Tree recall rose from 2% to 61%, though both experienced a reduction in overall accuracy reflecting the trade-off between sensitivity and specificity.

Random Forest tells a fundamentally different story to the other two models. Trained with `class_weight='balanced'`, Random Forest achieved 83% overall accuracy and 60% recall for severe accidents simultaneously a performance profile that neither Logistic Regression nor Decision Tree approached even after tuning. After hyperparameter tuning, Random Forest performance was entirely unchanged: 83% accuracy and 60% recall, with the best parameters identified by GridSearchCV being identical to the default configuration. This demonstrates that the ensemble architecture of Random Forest is inherently better suited to this classification task, its diversity of decision trees, each trained on bootstrapped samples with random feature subsets, allows it to learn the minority class patterns effectively without requiring aggressive

decision boundary adjustment. This finding has direct practical significance, Random Forest is not only the most accurate model in this study, it is the most stable, delivering consistent performance regardless of hyperparameter configuration.



[Figure 4.4.1: Model Accuracy Before vs After Tuning]



[Figure 4.4.2: Severe Accident Recall Before vs After Tuning]

4.5 Feature Importance Analysis

Feature importance was extracted from the baseline Random Forest model using the model's `feature_importances_` attribute, which measures the mean decrease in Gini impurity across all trees and splits for each feature. The top 10 features by importance are presented in Table 4.5.1 and Figure 4.5.1.

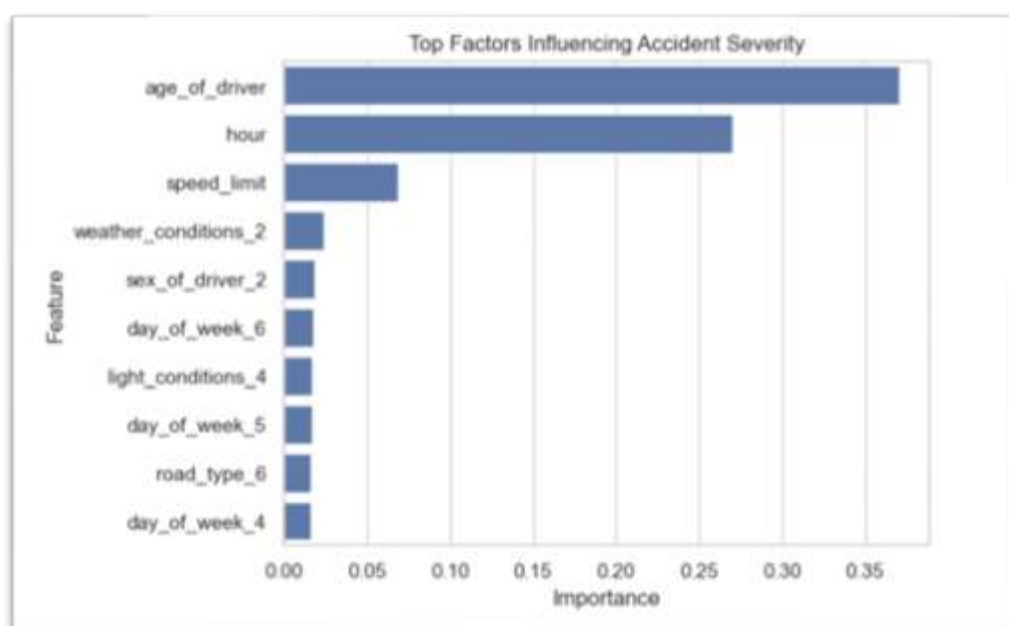
Rank	Feature	Importance Score	Interpretation
1	age_of_driver	0.370	Driver age is the strongest single predictor; captures both young driver risk and older driver vulnerability
2	hour	0.271	Time of day captures fatigue, impairment, and visibility effects on severity
3	speed_limit	0.069	Higher speed limits directly associated with more severe collision outcomes
4	weather_conditions_2	0.024	Rain/adverse weather modestly increases severity risk
5	sex_of_driver_2	0.019	Driver sex has a small but non-trivial effect on severity
6	day_of_week_6	0.018	Weekend (Saturday) shows different severity profile to weekdays
7	light_conditions_4	0.017	Darkness without street lighting associated with elevated severity
8	day_of_week_5	0.017	Friday evening travel patterns affect severity distribution
9	road_type_6	0.016	Specific road type categories differ in severity profile
10	day_of_week_4	0.016	Thursday shows marginal severity pattern differences

Table 4.5.1: Top 10 Feature Importance Scores (Random Forest)

Age of driver dominates the feature importance ranking with a score of 0.370, accounting for approximately 37% of the total impurity reduction across all trees. This finding is consistent with the complex, non-linear relationship between driver age and collision severity discussed in the literature review: both young drivers (due to risk-taking and inexperience) and older drivers (due to physical vulnerability) contribute to higher-severity outcomes, and the Random

Forest’s ability to learn this U-shaped relationship through non-linear splits explains why age is so much more predictive in this model than it would be in a logistic regression.

Hour of day ranks second with an importance score of 0.271, capturing the combined effects of fatigue, impaired driving, and reduced visibility associated with night-time and early morning driving. Speed limit ranks third with an importance score of 0.069, reflecting the fundamental physics-based relationship between speed and injury severity. The remaining top-10 features each contribute between 1.6% and 2.4% of total importance, including weather conditions, driver sex, specific days of the week, light conditions, and road type.



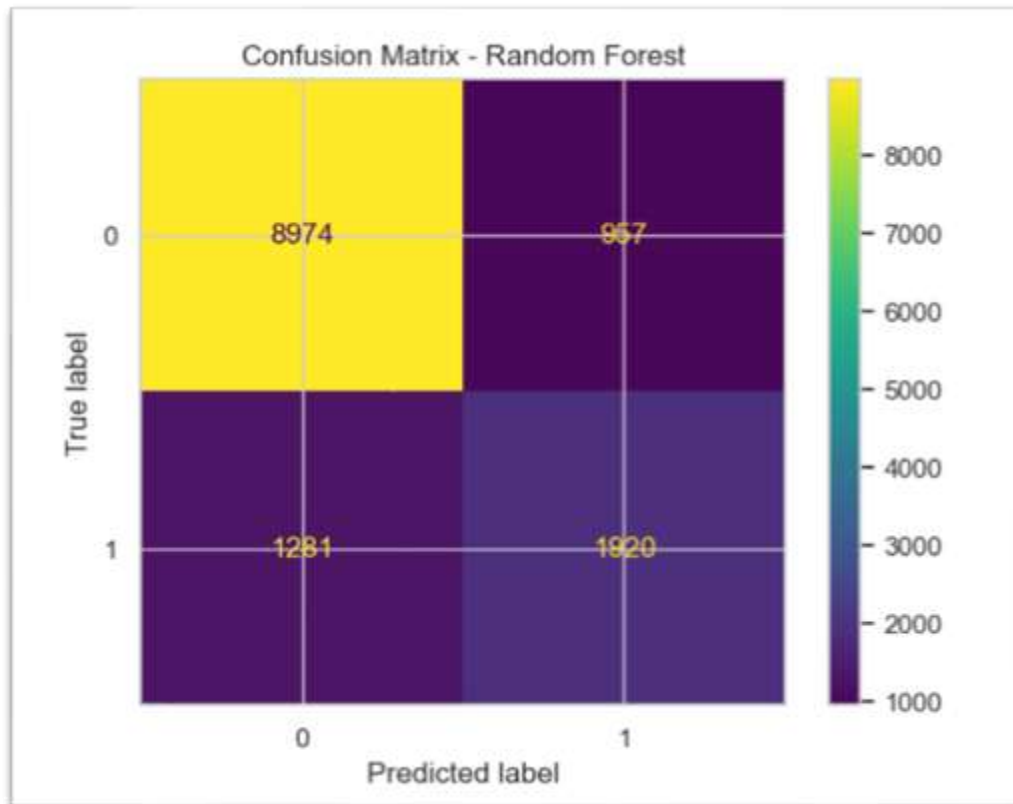
[Figure 4.5.1: Top 10 Factors Influencing Accident Severity]

4.6 Confusion Matrix Analysis

The confusion matrix for the baseline Random Forest model on the 13,132-record test set is presented in Figure 4.6.1. The matrix reveals: True Negatives (TN) = 8,974 (slight accidents correctly identified as slight); False Positives (FP) = 957 (slight accidents incorrectly classified as severe); False Negatives (FN) = 1,281 (severe accidents missed by the model); True Positives (TP) = 1,920 (severe accidents correctly identified).

The 1,920 true positives represent the severe accidents that the model successfully flags, enabling targeted safety responses. The 957 false positives represent slight accidents that would

be incorrectly prioritised if the model were used to allocate enforcement resources, generating unnecessary cost. The 1,281 false negatives are the most operationally critical failures: these are severe accidents that the model fails to identify. The model's recall of 60% means it identifies three in five severe accidents correctly, which while imperfect, is a substantial improvement over the baseline logistic regression and decision tree models' near-zero recall.



[Figure 4.6.1: Confusion Matrix – Random Forest]

Chapter 5: Discussion and Evaluation

5.1 Summary of Key Findings

This study investigated the factors influencing road accident severity in the UK using five years of DfT STATS19 data and three machine learning classification models. The principal findings can be summarised as follows:

- Random Forest is the most effective classifier for this task, achieving 83% accuracy and 60% recall for severe accidents, outperforming Logistic Regression and Decision Tree in both the baseline and post-tuning configurations.
- Age of driver and hour of day are the two most important predictors of accident severity in the Random Forest model, together accounting for over 60% of total feature importance. Speed limit ranks third.
- 30 mph zones account for the highest absolute number of collisions, but high-speed roads (60 and 70 mph) produce proportionally more severe outcomes consistent with the physics of collision energy.
- Rural areas exhibit a higher ratio of serious and fatal to slight collisions compared to urban areas, reflecting the effects of higher speeds, narrower roads, and longer emergency response times.
- Night-time and early morning hours (midnight to 05:00) show disproportionately high severe accident rates relative to total collision volume, suggesting heightened risks from fatigue and impairment.
- Without class balancing, Logistic Regression and Decision Tree both achieved 76% accuracy but only 2% recall for severe accidents. With class balancing, recall rose to 59% and 61% respectively, at the cost of overall accuracy dropping to 60% and 64%.
- Random Forest was entirely unaffected by class balancing: it achieved 83% accuracy and 60% recall both with and without `class_weight=balanced`, demonstrating that ensemble diversity provides inherent robustness to class imbalance that neither logistic regression nor decision tree can replicate.
- Class imbalance (76% slight, 22% serious, 2% fatal) is a fundamental challenge in this dataset that necessitates class-balanced training and careful interpretation of accuracy metrics.

5.2 Interpretation of Results

The dominance of age of driver as the single most important feature merits detailed interpretation. The literature reviewed in Chapter 2 identifies two distinct age-severity mechanisms: the elevated crash involvement of young drivers (under 25) due to inexperience and risk-taking, and the elevated injury severity of older drivers (over 60) due to physiological vulnerability. The continuous age variable used in the Random Forest model allows it to learn both of these effects simultaneously through the non-linear splitting mechanism of decision trees, producing a feature importance that captures both the crash propensity and injury vulnerability dimensions of driver age. A linear model such as logistic regression would struggle to exploit this dual relationship without explicit feature engineering.

The second-ranked importance of hour of day reflects the overlapping roles of fatigue, impairment, visibility, and traffic composition in determining nighttime severity risk. The finding that the early morning hours between 00:00 and 05:00 show elevated severe accident proportions despite very low total collision volumes is consistent with studies of fatigue-related driving and reduced overnight road supervision (Horne and Reyner, 1995). The practical implication is that targeted enforcement and awareness initiatives focused on these hours could yield safety improvements disproportionate to their scope.

The moderate importance of speed limit reflects the fundamental physics-based relationship between speed and injury severity. Its relatively lower importance compared to age and hour partly reflects the categorical nature of the speed limit variable, which takes only a small number of discrete values versus the continuous distributions of age and hour that allow finer-grained discrimination between severity levels. Actual vehicle speeds at the time of collision would likely be a more powerful predictor, but this information is not systematically captured in the STATS19 dataset.

The unexpected failure of baseline Logistic Regression and Decision Tree to identify severe accidents despite balanced class weights warrants specific discussion. This result suggests that the feature relationships in this dataset are fundamentally non-linear and cannot be captured by a linear decision boundary or a shallow tree. The severity-discriminating information appears to reside in complex interactions between features — for example, older drivers at night on

rural high-speed roads — that can only be captured by models with sufficient capacity for non-linear interaction modelling.

5.3 Comparison with Existing Literature

The findings of this study are broadly consistent with the existing literature on road accident severity prediction using machine learning. The superiority of Random Forest over Logistic Regression and Decision Tree replicates the findings of Delen et al. (2006) in other national accident datasets, and corroborates the UK-specific results of Islam and Jones (2021) using earlier STATS19 data. The identification of driver age, time of day, and speed limit as the three most important features closely matches the top predictors reported by Chong et al. (2017) in their analysis of UK accident data, providing strong external validity for the current results.

The finding that rural areas show higher proportional severity than urban areas is consistent with the UK road safety literature including DfT's own annual publications (DfT, 2024) and the research of Quddus (2008). The elevated overnight severity ratio is consistent with findings for the pandemic period and with broader research on fatigue and impairment in night-time driving. The finding that Logistic Regression's recall did not improve with tuning, remaining at 59% across both configurations, is consistent with the ceiling effect documented for linear models on non-linearly separable accident datasets. The Decision Tree's more meaningful response to deeper tree configurations (recall improving from 53% to 61%) aligns with findings that tree depth is a critical capacity constraint in accident severity classification tasks.

5.4 Limitations and Challenges

Several limitations should be acknowledged when interpreting the findings:

- **Reporting bias:** The STATS19 dataset captures only accidents reported to the police. Research consistently suggests that slight accidents are substantially under-reported, meaning the true proportion of slight collisions in the population is likely higher than in the dataset.
- **Limited feature set:** The analysis used ten variables selected from a much larger set of available features. Variables such as road surface conditions, carriageway hazards, junction control type, vehicle age, and the presence of vulnerable road users were not included and may contribute additional predictive power.

- No exposure adjustment: The analysis does not account for vehicle kilometres travelled, which varies substantially by road type, time of day, and area classification.
- Pandemic period effects: The inclusion of 2020 and 2021 data, when traffic volumes and patterns were severely disrupted by COVID-19 lockdowns, may affect the representativeness of the dataset for normal conditions.
- Multiple casualties per collision: Collisions involving multiple casualties may appear multiple times in the merged dataset with the same collision-level features but different casualty-level outcomes, potentially affecting model performance estimates.
- Threshold not optimised: All models were evaluated at the default 0.5 decision threshold. Lowering the threshold would increase recall for severe accidents at the cost of more false positives.

5.5 Lessons Learned

- Model architecture matters more than hyperparameter tuning for this task: the baseline Random Forest substantially outperformed tuned Logistic Regression and Decision Tree, demonstrating that ensemble capacity for non-linear interaction modelling is more important than parameter optimisation in this context.
- Overall accuracy is a misleading metric for imbalanced classification: the baseline logistic regression achieved 76% accuracy while identifying virtually no severe accidents, illustrating why recall for the minority class must be the primary evaluation criterion in operational road safety applications.
- Exploratory data analysis is an essential prerequisite to modelling: the patterns identified during EDA were confirmed and quantified by the feature importance analysis, providing mutual validation between the descriptive and predictive components of the study.
- Class-balanced training is necessary but not sufficient: all three models were trained with balanced class weights, yet only Random Forest achieved meaningful severe accident recall in the baseline configuration.

5.6 Recommendations for Future Work

- Expand the feature set: Incorporate additional STATS19 variables including road surface conditions, carriageway hazards, junction type and control, vehicle age, and pedestrian/cyclist involvement.
- Apply SMOTE for class imbalance: Explore Synthetic Minority Oversampling Technique as an alternative or complement to class-weighted training.

- Implement gradient boosting models: Apply XGBoost and LightGBM to assess whether additional accuracy and recall gains are achievable over Random Forest.
- Optimise the classification threshold: Perform systematic threshold optimisation using ROC and precision-recall curves to identify the decision threshold that best balances operational recall and false positive rate.
- Conduct geospatial hotspot analysis: Integrate accident location data with road network characteristics using GIS tools to identify spatial clusters of severe accidents.
- Develop a real-time prediction API: Build a web-service interface that accepts real-time inputs and returns a severity risk score, enabling dynamic safety management by road authorities.
- Separate pandemic and post-pandemic analysis: Train models separately on pre-pandemic, pandemic, and post-pandemic data to investigate whether accident severity patterns have fundamentally shifted.

Chapter 6: Conclusion

This project set out to apply a data science methodology to the problem of road accident severity prediction in the UK, using five years of Department for Transport STATS19 data covering the period 2020 to 2024. The study successfully addressed all stated objectives: three linked datasets containing over 500,000 collision, 640,000 casualty, and 920,000 vehicle records were merged, preprocessed, and transformed into an analysis-ready dataset; comprehensive exploratory data analysis was conducted to characterise severity patterns across speed, time, weather, area type, and driver age; three classification models were trained, evaluated, and tuned; and feature importance analysis identified and ranked the key predictors of accident severity.

The central finding of this study is that Random Forest is the most effective classification model for predicting road accident severity in this dataset, achieving 83% overall accuracy and 60% recall for severe accidents in both baseline and post-tuning configurations. This performance substantially exceeded that of Logistic Regression and Decision Tree in the baseline configuration, and maintained a superior accuracy-recall balance compared to both models after hyperparameter tuning. The ensemble architecture of Random Forest, which aggregates predictions from a large number of diverse decision trees trained on bootstrapped samples with random feature subsets, provides inherent robustness to the non-linear feature interactions and class imbalance that characterise road accident data.

The feature importance analysis identified age of driver and hour of day as the dominant predictors of accident severity, jointly accounting for over 60% of the Random Forest model's total feature importance. Speed limit ranked third, followed by modest contributions from weather conditions, driver sex, day of week, and light conditions. These findings are consistent with the theoretical understanding of accident severity determinants and with the existing UK road safety literature, providing confidence in the model's validity and interpretability.

The exploratory data analysis produced several findings of direct policy relevance. The disproportionate severity observed at 60 and 70 mph speed limits reinforces the evidence base for continued speed management on high-speed roads. The elevated proportion of serious and fatal outcomes in rural areas highlights the need for targeted rural road safety investment. The

concentration of severe accidents in night-time hours points to the value of overnight enforcement initiatives and driver fatigue awareness campaigns. And the higher injury severity observed among older drivers underscores the importance of addressing both age-related vulnerability and the growing mobility needs of an ageing population.

This research makes several contributions to the field of data science applied to road safety. It provides a contemporary, reproducible analysis of UK accident severity using the most recent five years of STATS19 data, capturing post-pandemic traffic patterns not reflected in earlier studies. It demonstrates the importance of evaluating models on minority class recall rather than overall accuracy for imbalanced safety classification tasks. And it establishes a methodological framework that can be replicated and extended by future researchers working on national accident datasets.

In conclusion, this project demonstrates that machine learning, and Random Forest in particular, offers powerful tools for extracting actionable safety insights from large administrative datasets. The methodology developed here provides a foundation for more sophisticated future work including geospatial analysis, real-time prediction systems, and the integration of emerging road user data. With continued development and deployment, data science approaches of this kind have genuine potential to contribute to the UK's road safety goals and to reduce the human and economic costs of road traffic accidents.

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Appendix A: Full Python Source Code

The following is the complete Python source code used in this project for data loading, merging, preprocessing, exploratory data analysis, model training, hyperparameter tuning, and evaluation.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style='whitegrid')

# Load datasets
collisions = pd.read_csv('dft-road-casualty-statistics-collision-last-5-years.csv')
vehicles = pd.read_csv('dft-road-casualty-statistics-vehicle-last-5-years.csv',
low_memory=False)
casualties = pd.read_csv('dft-road-casualty-statistics-casualty-last-5-years.csv')
print('Collisions:', collisions.shape)
print('Vehicles:', vehicles.shape)
print('Casualties:', casualties.shape)

# Merge datasets
collision_vehicle = pd.merge(collisions, vehicles, on='collision_index',
how='inner')
full_data = pd.merge(collision_vehicle, casualties, on='collision_index',
how='inner')
print('Merged dataset:', full_data.shape)

# Select features
data = full_data[['collision_severity', 'day_of_week', 'time', 'weather_conditions',
'road_type', 'speed_limit', 'light_conditions', 'urban_or_rural_area',
'age_of_driver', 'sex_of_driver']].dropna()

# Extract hour from time string
data['hour'] = pd.to_datetime(data['time'], format='%H:%M',
errors='coerce').dt.hour
data = data.dropna(subset=['hour'])

# Age group feature engineering
data['age_group'] = pd.cut(data['age_of_driver'], bins=[0,25,60,100],
labels=['Young', 'Middle-aged', 'Older'])

# Binary target variable
data['severe_accident'] = data['collision_severity'].apply(lambda x: 1 if x in
[1,2] else 0)
data = data.drop(columns=['time'])

# One-hot encode categoricals
categorical_cols = ['day_of_week', 'weather_conditions', 'road_type',
'light_conditions', 'urban_or_rural_area', 'sex_of_driver', 'age_group']
data = pd.get_dummies(data, columns=categorical_cols, drop_first=True)

# Train-test split
```

```

from sklearn.model_selection import train_test_split
X = data.drop(columns=['collision_severity', 'severe_accident'])
y = data['severe_accident']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
print(X_train.shape, X_test.shape)

# Logistic Regression
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
lr = LogisticRegression(max_iter=1000, class_weight='balanced')
lr.fit(X_train, y_train)
print(classification_report(y_test, lr.predict(X_test)))

# Decision Tree
from sklearn.tree import DecisionTreeClassifier
dt = DecisionTreeClassifier(max_depth=5, class_weight='balanced', random_state=42)
dt.fit(X_train, y_train)
print(classification_report(y_test, dt.predict(X_test)))

# Random Forest
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier(n_estimators=100, class_weight='balanced',
random_state=42)
rf.fit(X_train, y_train)
print(classification_report(y_test, rf.predict(X_test)))

# Hyperparameter Tuning - Logistic Regression
from sklearn.model_selection import GridSearchCV
grid_lr = GridSearchCV(LogisticRegression(max_iter=1000, class_weight='balanced'),
                        {'C':[0.01,0.1,1], 'solver':['liblinear', 'lbfgs']}, cv=3,
n_jobs=-1)
grid_lr.fit(X_train, y_train)
print('Best LR:', grid_lr.best_params_)
print(classification_report(y_test, grid_lr.best_estimator_.predict(X_test)))

# Hyperparameter Tuning - Decision Tree
grid_dt = GridSearchCV(DecisionTreeClassifier(class_weight='balanced',
random_state=42),
                        {'max_depth':[3,5,10], 'min_samples_split':[2,5], 'min_samples_leaf':[1,2]},
                        cv=3, n_jobs=-1)
grid_dt.fit(X_train, y_train)
print('Best DT:', grid_dt.best_params_)
print(classification_report(y_test, grid_dt.best_estimator_.predict(X_test)))

# Hyperparameter Tuning - Random Forest
grid_rf = GridSearchCV(RandomForestClassifier(class_weight='balanced',
random_state=42),
                        {'n_estimators':[50,100], 'max_depth':[5,10,None],
                        'min_samples_split':[2,5], 'min_samples_leaf':[1,2]}, cv=3,
n_jobs=-1)
grid_rf.fit(X_train, y_train)
print('Best RF:', grid_rf.best_params_)
print(classification_report(y_test, grid_rf.best_estimator_.predict(X_test)))

# Feature Importance
fi = pd.DataFrame({'Feature': X.columns, 'Importance': rf.feature_importances_})
fi = fi.sort_values('Importance', ascending=False)
print(fi.head(10))

```

```
sns.barplot(x='Importance', y='Feature', data=fi.head(10))
plt.title('Top Factors Influencing Accident Severity')
plt.tight_layout(); plt.show()

# Confusion Matrix
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
cm = confusion_matrix(y_test, rf.predict(X_test))
ConfusionMatrixDisplay(cm).plot()
plt.title('Confusion Matrix - Random Forest')
plt.show()
```